How Wrong Can Your Simulator Be? Auditing, Tolerating, and Updating a Measured Forward Model on a Real Instrument

Anonymous Author(s)
Affiliation
Address
email

Abstract

A learned restoration model for degraded macro X-ray fluorescence (MA-XRF) scans of a painting is trained on a forward simulator whose every ingredient (geometry, per-line gains, noise law) is measured on the instrument; we treat the simulator's imperfection as the object of study. An ensemble trained across simulators drawn within the measured calibration uncertainty separates simulator uncertainty from initialisation variance (2.8× wider than a fixed-simulator control on the real scan, 86 % of its variance attributable to the simulator), carries a predictive band far closer to calibrated on the real scan than the control's (0.61/0.87/0.95 vs 0.53/0.80/0.91 at 1/2/3σ; both still below the Gaussian 0.68/0.95/0.997), and its mean is the first learned restoration in our study that does not lose correlation to the physics inverse on the real scan. Eighteen deliberately broken training simulators show the learned refinement tolerates large noise and gain errors but fails on geometry: 0.5 px of registration error already hurts, 1 px makes the network worse than the physics it refines. A blind diagnostic battery thresholded on the calibration uncertainty identifies every defect that causes material damage and, on the real scan, rediscovers unsupervised the resampling-blur mismatch that originally broke our training pipeline. Used as ABC summaries, the same statistics turn one real scan into a posterior over the simulator: the blur defect is rejected (prior 0.5, posterior 0.000) and a posterior-trained ensemble is the study's most accurate restoration, with a 27 % narrower band than the prior's. A 180-case degradation grid with four classical inpainting controls shows where learning pays: everywhere on measured pixels, nowhere inside missing regions, where a biharmonic fill wins in 16 of 16 cells. All results are anchored on real measurements from one instrument and one painting; all negative results are reported.

1 Introduction

Machine learning for experimental science usually trains on simulations because ground truth is scarce, and the simulator is never exactly the instrument; the common response is to validate it once and trust it. This paper takes the opposite view on a small but fully measured problem: we ask *how* imperfect a validated simulator still is, what that costs a network trained on it, whether it can be detected and attributed blind, how to represent it in predictive uncertainty, and whether one real measurement can reduce it.

Our testbed is the restoration of geometrically degraded MA-XRF element maps of an oil painting [1]: a scan acquired at a tilt must be registered, gain-corrected and denoised back into the frontal frame. The forward model is not designed but measured (Section 2), which gives us something rare: a per-component calibration uncertainty for every knob of the simulator. Our contributions:

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

- **A documented failure mode of simulator validation.** A forward model can pass a fidelity gate as an emulator and still be wrong as a training-data generator; we document the mechanism (a single resampling blur, Section 2) and reuse it as a controlled defect, diagnostic target and ABC hypothesis.
- **Uncertainty that knows the simulator is imperfect.** Deep ensembles [8] trained across simulator draws within the measured calibration uncertainty, against a fixed-simulator control that isolates initialisation variance (Section 3): domain randomization [11] with measured ranges.
- **A defect-tolerance ladder and a blind audit.** One network per broken simulator quantifies which defects hurt; a summary-statistic battery thresholded on the calibration uncertainty identifies which component is broken without ground truth (Section 4); every defect with material damage is visible to the audit.
- **Closing the loop.** The audit statistics are ABC summaries [2, 4]: one real scan yields a posterior over the simulator's knobs, a posterior-trained ensemble, and a posterior predictive check that names precisely what the simulator still lacks (Section 5).
- **A regime map against classical baselines.** A 180-case grid over tilt, missing data and dose, with four classical inpainting controls, locates where learning pays (Section 6); inside missing regions the network loses to a classical fill.

2 Setup: a measured forward model and its failure mode

Instrument and data. Two silicon drift detectors (70 mm^2 mounted horizontally and 25 mm^2 at 40° from horizontal, 12.5 µm Be windows) and a tube at 37 kV / 40 µA scan an oil painting; per-pixel spectra are fitted into net-count maps of 8 element lines (Ca, Ti, Fe, Cu, four Pb L), summed over the detectors. Three scans of the *same painting*: two frontal, F_1 and F_2 (60 × 120 px, seven days apart, same geometry), and one real tilted, T (45 × 80, mounted at 7.7°). F_1 is the only training source; F_2 and T are never seen in training.

Forward model. Every ingredient is measured, not modelled: the warp is the affine registration fitted between T and F_1; the per-line tilt gains are the measured level response at the reference mounting, extrapolated linearly in angle; the noise is Poisson-consistent with a per-line constant, $\mathrm{Var}(m) = k_{\mathrm{el}} m$ with $k_{\mathrm{el}} = 4.2$–9.0 calibrated from the F_1–F_2 difference. A fidelity gate holds on all 8 lines: $r(\mathrm{sim,T}) = 0.89$–0.98, within 0.02 of the F_1–F_2 repeatability ceiling, and levels within 0.7 %. A residual U-Net [9] (469k parameters, zero-initialised head, so the untrained network *is* the physics inverse) refines the deterministic inversion; training pairs are simulated from F_1 at random angles, doses and dropout blocks (Appendix A).

The blur lesson. Our first training simulator emulated the tilted acquisition by bilinearly warping the frontal map. It passed the fidelity gate, yet the network trained on it *over-sharpened* the real scan (contrast ratio up to 1.05, correlation below the physics inverse on 8/8 lines). A real tilted scan is a direct measurement with no resampling blur, so the simulated restoration inputs carried two interpolation blurs where reality has one. Sampling the source with a cubic spline at the exact warp positions (v2) fixed the training distribution. The transferable lesson: *validated as an emulator is not validated as a training-data generator*, and the failure was invisible on simulated tests.

Perturbable simulator. We expose the simulator's components as knobs θ: a noise-constant scale, a tilt-gain slope scale, per-line gain offsets, the assumed mounting angle, registration shift and rotation, the interpolation order (cubic vs the v1 bilinear), and, in Section 5, per-line noise scales. Each knob carries a measured 1σ calibration uncertainty (noise log-sd 0.33 from the session transfer of k; gain slope 20 %, the mounting angle being an upper bound; registration 0.1 px, 0.3° from the spread of per-element fits). Perturbations apply only to the training-data simulator; the test-time inverse stays nominal, as in practice.

3 Uncertainty that knows the simulator is imperfect

We train two ensembles of $N{=}12$ members each: a *jitter* ensemble, member i trained on a simulator drawn from the calibration uncertainty, and a *control* ensemble with the nominal simulator and the

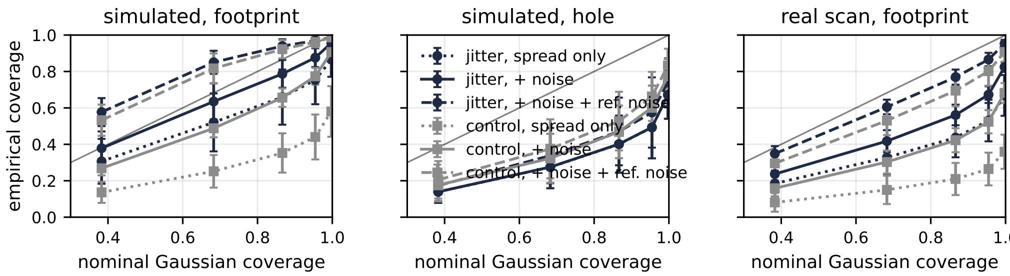

Figure 1: Coverage of $|\mathrm{mean} - \mathrm{truth}| \leq z\sigma$ vs the Gaussian target, jitter (navy) and control (grey); dotted: spread only, solid: plus propagated noise, dashed: plus reference-map noise (sim upper bound; the correct band on the real scan). Bars: sd over lines and cases. Left: 32 simulated cases (footprint); middle: in a 14×20 hole; right: the real scan.

same twelve seeds, the standard deep-ensemble baseline [8]. By construction the excess spread of the jitter ensemble is the variance attributable to simulator uncertainty. The predictive band adds measurement noise propagated through 8 re-simulations per case plus, on the real scan, the calibrated noise of the reference map ($k_{\mathrm{el}} \cdot \mathrm{truth}$), since the reference is itself a measurement.

The ensemble knows where the simulator is uncertain. On the footprint the jitter ensemble spreads $2.2\times$ (simulated) and $2.8\times$ (real scan) wider than the control; 62 % and 86 % of its variance is attributable to the simulator draw. The simulator-attributable spread (rms over footprint pixels) grows monotonically with tilt, from 104 counts at 8° to 148 at 25°, the direction of the gain extrapolation from the single 7.7° calibration; spatially it follows edges and levels while the control spread lives only in dropout holes (Appendix Fig. 5).

Calibration. With spread alone both ensembles under-cover (real scan, z=2: 0.53 jitter, 0.26 control). With the full band the jitter ensemble covers 0.61/0.87/0.95 at $z = 1/2/3$ on the real scan against 0.53/0.80/0.91 for the control (Fig. 1); on simulated cases it is bracketed between 0.64 and 0.85 at z=1 because the reference maps are themselves noisy. This is still under-coverage with heavy tails (3σ failure rate 0.047 vs nominal 0.003), not calibration; our claim is the difference to the control, the simulator's share. The jitter spread also ranks the real-scan errors better (Spearman 0.31 vs 0.26, AUSE [6] 0.33 vs 0.38).

Where it fails. Inside a 14×20 dropout hole neither ensemble is calibrated (z=2 coverage 0.49/0.60 against rms error 940 counts vs spread 320–410), the spread does not rank the error (Spearman $\approx$ 0.1), and the jitter ensemble is *narrower* than the control (ratio 0.85): all members share the same image prior and agree on the same wrong fill; simulator uncertainty is not prior uncertainty.

Jitter training costs nothing on the real scan. The jitter-ensemble mean matches or exceeds the physics inverse in r on 5/8 lines (mean 0.9527 vs 0.9526, the other lines within 0.002) while restoring contrast (cv ratio 0.979 vs 0.947), whereas the single nominal network and the control mean lose r on 8/8 lines (0.9485, 0.9496). Calibration-posterior randomization is a physically grounded augmentation against the real scan's residual mismatch, at a price of 0.003 in r on simulated cases, where the nominal simulator is true.

4 Auditing the simulator: tolerance and blind diagnostics

Tolerance ladder. One network per broken simulator, 18 rungs across six families, one shared seed so that rung differences are the simulator; the 12 control members give the seed band ($\Delta r \in [+0.082, +0.090]$ simulated footprint, $[-0.005, -0.002]$ real). The result (Fig. 2) is asymmetric: the noise constant can be wrong by $4\times$ either way, the gain slope doubled or removed, the angle belief off by 5°, all without leaving the band. Geometry differs: a 0.5 px shift, five times the calibration uncertainty yet half a pixel, already leaves the band (four seeds: -0.013 to -0.005 real); at 1 px the network is worse than the physics inverse it refines (-0.057 simulated, -0.086 real), at 2 px it destroys the restoration (-0.27 real); a 2° rotation costs -0.047. The blur rung replays Section 2:

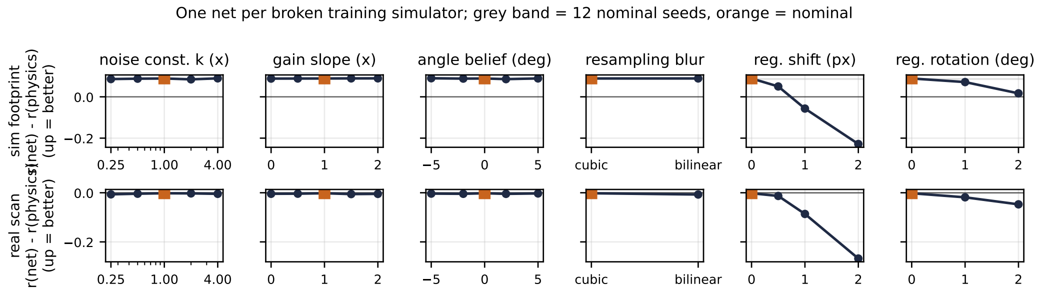

Figure 2: Defect tolerance: $r(\text{net}) - r(\text{physics})$ (up = better), one network per broken training simulator, on nominal simulated cases (top) and the real scan (bottom); grey band: 12 nominal seeds; orange: nominal. The hole region is omitted (seed band $[-0.03, +0.27]$, no tolerance signal). Geometry is the cliff; blur is visible only on the real scan.

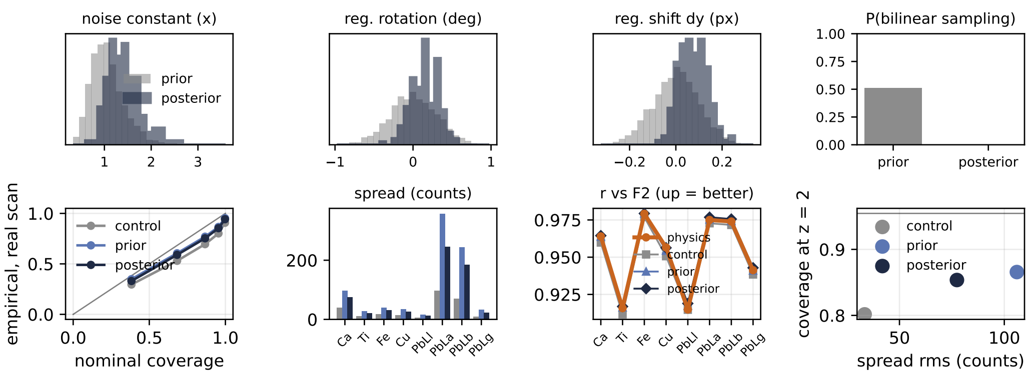

Figure 3: Top: prior (grey) and posterior (navy) ABC marginals from one real scan, for the knobs the data moves (gain slope and angle match the prior; omitted); P(bilinear) drops from 0.5 to 0.000. Bottom: real-scan calibration, per-line spread and r (physics in orange), and the coverage-vs-width tradeoff.

invisible on simulated tests, it leaves the band on the real scan for all four seeds, with the table's only contrast overshoot (1.013).

Blind diagnostics. Can the broken component be identified from one scan, without truth? We compare measurement to simulation with nine per-line statistics (level, contrast, mid- and high-band power ratios, a noise-constant estimate, signed sub-pixel offsets, a rotation estimate, a gain-template projection). The null is the crucial choice: with noise-only pairs everything fires on the real scan (a 0.3° registration residual gives $z \approx 60$), so we build it from pairs drawn *within the calibration uncertainty*: a flag means the discrepancy exceeds what the simulator claims to know. With a rule fixed before the experiment, the battery identifies blur 10/10, shifts down to 0.5 px 30/30, rotations of 1–2° 20/20 and noise scalings $\times$0.25/2/4 30/30, with 4/48 false alarms; gain-family defects within the calibration uncertainty are invisible by construction (0/80; a post-hoc template recovers only the extremes). Crossing audit and ladder gives the payoff matrix: *every rung with material damage is visible, and every invisible rung is harmless* (one formal exception at 0.0002, the band's own resolution). On the real scan the battery, blind, flags our v1 simulator and the validated bilinear emulator as "blur" (high-frequency power ratio 1.7–2.1) and leaves the v2 simulator with one finding, a residual noise excess (k ratio 1.19): it rediscovers the blur lesson (Appendix Fig. 7).

5 Closing the loop: a posterior over the simulator from one scan

The audit statistics support likelihood-free inference [4]: rejection ABC [2] over the simulator's knobs, prior = the calibration jitter of Section 3 plus the v1 bilinear sampling at probability 0.5, so

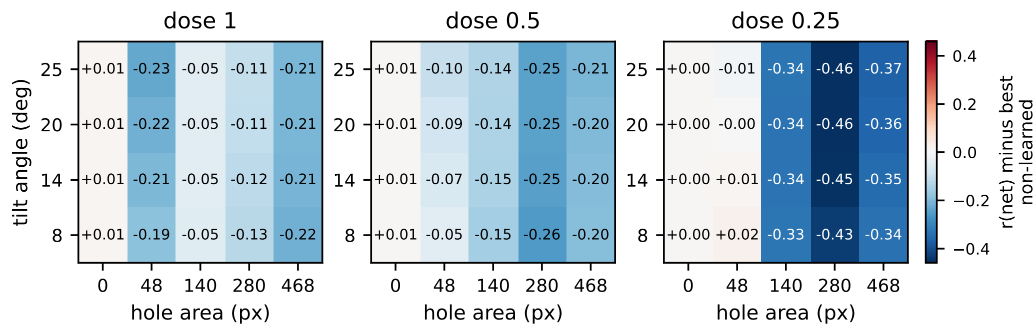

Figure 4: Regime map: $r(\mathrm{net})$ minus the best non-learned candidate (physics or best classical fill) inside the hole (footprint for the no-hole column), per dose; blue = the non-learned candidate wins, red = the network wins. Per-candidate curves in Appendix Fig. 8.

the posterior must reject the v1 defect on its own; 3000 draws, whitened RMS-z distance, closest 5 % accepted (2 % and 10 % agree in every direction).

Posterior. The bilinear defect is rejected (posterior 0.000); the posterior concentrates on a noisier instrument (noise scale 1.43 ± 0.35) with a small registration residual ($+0.15° \pm 0.17$, $+0.08 \pm 0.06$ px), gains at calibration. The posterior predictive check rejects every draw (0/12 consistent), with a uniform "noise" verdict whose decomposition shows a *per-line* residual (variance ratios 0.9–4.6 across lines) that a global noise knob cannot express. Adding the named knob (round 2) moves the marginals in the measured direction but barely deepens the fit (best whitened distance 2.02 vs a null floor of 0.95 ± 0.24): what remains is outside the knob family (unmodelled flat-field, scatter, dwell drift).

Posterior ensemble. Twelve networks retrained on accepted draws form the most accurate restoration in the study on the real scan: $r = 0.9539$ against 0.9527 (prior), 0.9526 (physics), 0.9496 (control), at or above the physics inverse on 7/8 lines, with the smallest bias (1.07 %) and a 27 % narrower band than the prior ensemble (spread 77 vs 106 counts) at coverage 0.85/0.94 at $z = 2/3$ (prior 0.87/0.95). By per-pixel Gaussian NLL, a proper score [5], the wider prior band remains marginally better (6.45 vs 6.58 nats; control 7.27), the gap sitting on the lines whose noise the knob family cannot express: one measurement bought accuracy, bias and band width; the score prices what remains.

6 When does learning pay? A regime map with classical controls

We sweep 180 cases (4 angles $\times$ 5 holes $\times$ 3 doses $\times$ 3 seeds, all from F_2) and compare physics, the nominal network, the jitter-ensemble mean, and four classical fills in the tilted frame (nearest, biharmonic [12], Telea [10], Navier–Stokes [3]), all through the same physics inverse. On measured pixels the learned candidates beat every alternative in every cell (no hole: 0.985 vs 0.977 physics; 14×20 hole footprint: 0.952 ensemble vs 0.937 best classical vs 0.781 physics), with a margin that grows with hole size. Inside the hole the ranking inverts: the biharmonic fill beats the network in 16/16 cells (0.467 vs 0.415 at $20°/14 \times 20$), and at quarter dose the learned fills collapse (0.03 and -0.01) while the classical fill is dose-proof (0.455). The only learned candidate that wins every hole cell is the hybrid, classical fill then network, by 0.005–0.03: the network adds sharpening, not a better fill. Hole columns are angle-independent, since the measured warp does not scale with angle. On this instrument the learned prior buys contrast, noise, neighbourhood repair and calibrated uncertainty, not missing content (Fig. 4).

Limitations and outlook. All of this is one painting, one instrument, one real tilted scan; we claim the auditable chain, not universal numbers. The calibration sigmas are lower bounds (the one-sided gain-slope uncertainty is symmetrised) and the jitter only as honest as they are; ensembles have 12 members, rungs one seed (borderline rungs four); rejection ABC in 22 round-2 dimensions is at its resolution edge (SMC is the upgrade). The hole failure suggests training on classically filled holes; uncertainty-guided acquisition (Appendix Fig. 9) is the route to less irradiation of heritage objects, our main foreseen societal benefit; code and result tables will be released.

References

[1] M. Alfeld, J. V. Pedroso, M. van Eikema Hommes, G. Van der Snickt, G. Tauber, J. Blaas, M. Haschke, K. Erler, J. Dik, and K. Janssens. A mobile instrument for in situ scanning macro-XRF investigation of historical paintings. *Journal of Analytical Atomic Spectrometry*, 28(5):760–767, 2013.

[2] M. A. Beaumont, W. Zhang, and D. J. Balding. Approximate Bayesian computation in population genetics. *Genetics*, 162(4):2025–2035, 2002.

[3] M. Bertalmio, A. L. Bertozzi, and G. Sapiro. Navier–Stokes, fluid dynamics, and image and video inpainting. In *IEEE CVPR*, 2001.

[4] K. Cranmer, J. Brehmer, and G. Louppe. The frontier of simulation-based inference. *PNAS*, 117(48):30055–30062, 2020.

[5] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[6] E. Ilg, Ö. Çiçek, S. Galesso, A. Klein, O. Makansi, F. Hutter, and T. Brox. Uncertainty estimates and multi-hypotheses networks for optical flow. In *ECCV*, 2018.

[7] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. In *ICLR*, 2015.

[8] B. Lakshminarayanan, A. Pritzel, and C. Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *NeurIPS*, 2017.

[9] O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional networks for biomedical image segmentation. In *MICCAI*, 2015.

[10] A. Telea. An image inpainting technique based on the fast marching method. *Journal of Graphics Tools*, 9(1):23–34, 2004.

[11] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IEEE/RSJ IROS*, 2017.

[12] S. van der Walt, J. L. Schönberger, J. Nunez-Iglesias, F. Boulogne, J. D. Warner, N. Yager, E. Gouillart, and T. Yu. scikit-image: image processing in Python. *PeerJ*, 2:e453, 2014.

A Training and evaluation details

Data and protocol. Frontal maps are 60×120, the tilted frame 45×80; the tilted footprint covers 3466 of the 7200 frontal pixels. Training pairs are simulated from F_1 only: random tilt angle in $[4°, 25°]$, random flips, 25 % low-dose samples (×0.35–1.0), 30 % of samples with 1–2 zeroed dropout blocks of 8–17 px in the tilted frame, recorded in a validity channel. The network input is the deterministic physics inverse of the simulated tilted scan (8 element channels, per-line normalisation fixed from F_1), plus validity and angle channels; the target is the frontal map. Four fixed 15×30 spatial blocks of the frontal frame are excluded from the training loss and serve early stopping. F_2 and the real scan T are never used in training; all final numbers are computed against F_2 as reference, on the footprint and, where applicable, inside the dropout hole. Metrics: Pearson r, SSIM, level bias in percent, and the contrast ratio $\mathrm{CV}(\mathrm{candidate})/\mathrm{CV}(\mathrm{reference})$.

Architecture and optimisation. Residual U-Net, three resolution levels, base width 32, GroupNorm and SiLU, 469k parameters, 1×1 zero-initialised head (the untrained model equals the physics inverse). Adam [7] with learning rate 10^{-3}, batch 8, at most 1500 steps, validation every 50 steps on a fixed simulated set (8 angles × 3 replicates), early stopping after 10 validations without improvement, plateau learning-rate decay. One training run takes about 10 minutes on a 12-core CPU workstation; no GPUs were used anywhere. The study trains 60 networks in total (12 jitter, 12 control, 18 ladder rungs, 6 extra-seed variants, 12 posterior members), about 10 CPU-hours, plus inference-only sweeps.

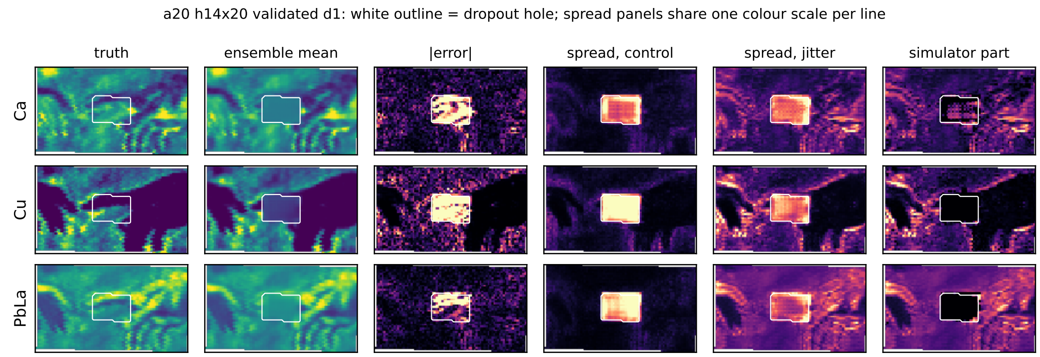

Figure 5: Spread maps on the harsh case (20°, 14×20 hole, white outline), three lines. Columns: reference, jitter-ensemble mean, absolute error, control spread, jitter spread, and the simulator part $\sqrt{\max(\sigma^2_{\text{jitter}} - \sigma^2_{\text{control}}, 0)}$; the three spread panels of a line share one colour scale. The control spread lives in the hole; the jitter spread follows the painting's structure; inside the hole the jitter ensemble is narrower than the control (dark).

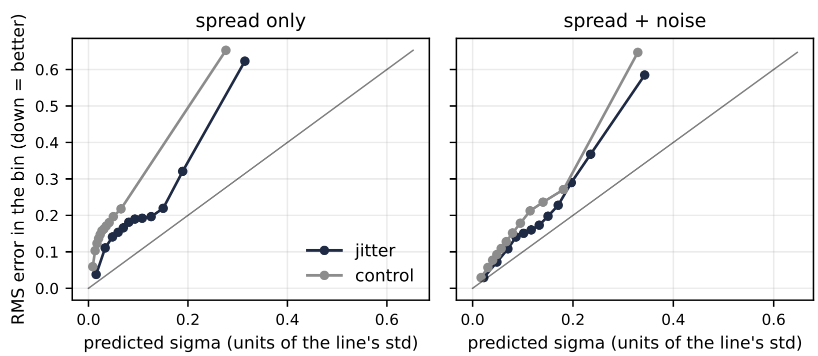

Figure 6: Binned reliability: RMS error against predicted σ in twelve σ-quantile bins, pooled over simulated cases and lines in units of each line's standard deviation; diagonal = calibrated. Left: ensemble spread only; right: spread plus propagated noise.

Seeds and error bars. The two ensembles share their twelve seeds, so jitter-vs-control differences are attributable to the simulator draw; every ladder rung uses the seed of one control member, and the 12 control members define the initialisation band shown in Fig. 2; the two borderline rungs (0.5 px shift, blur) were re-trained with three extra seeds each and their real-scan damage holds for every seed. Coverage error bars are standard deviations over lines and cases. The aleatoric band term uses 8 noise re-simulations per case; the reference-noise term uses the calibrated k_{el}.

B Additional figures

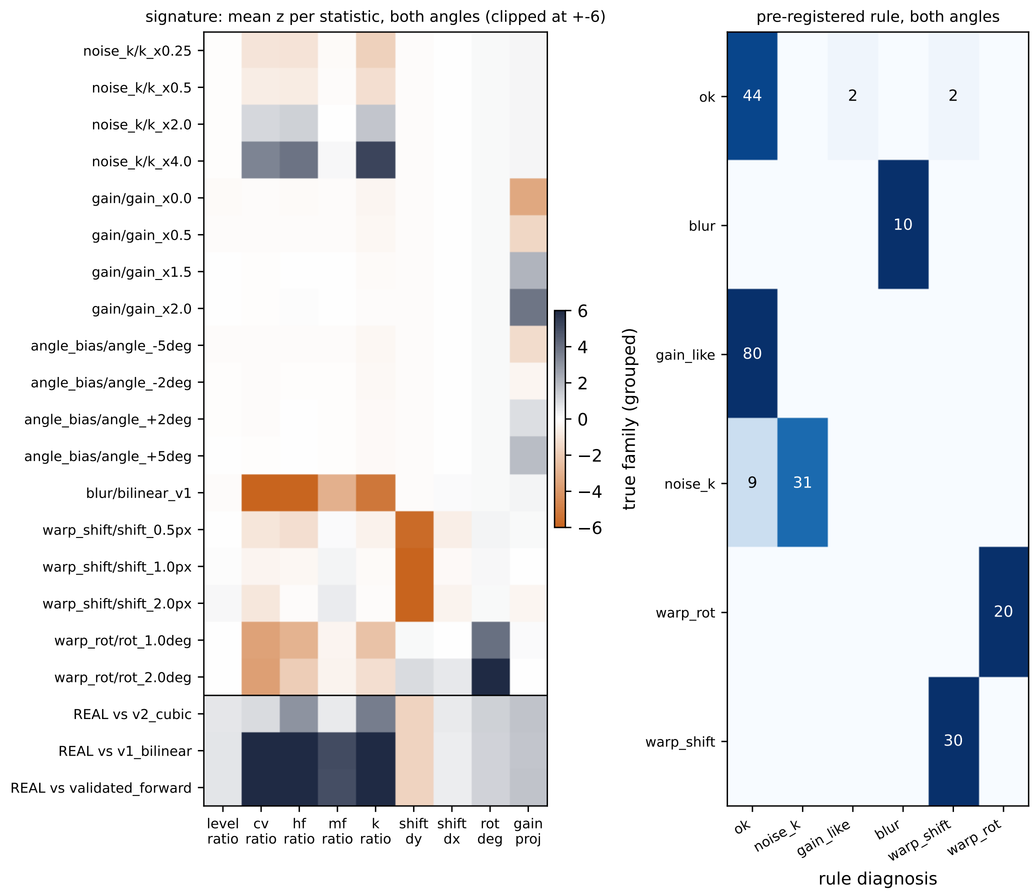

Figure 7: Blind diagnostics. Left: signature of every defect rung, the mean z of each statistic against the calibration-uncertainty null (both test angles pooled, clipped at ± 6); the bottom block is the real scan against three simulators (v2 cubic, v1 bilinear, validated bilinear emulator). Right: identification counts of the pre-registered rule, true family (grouped) versus diagnosis.

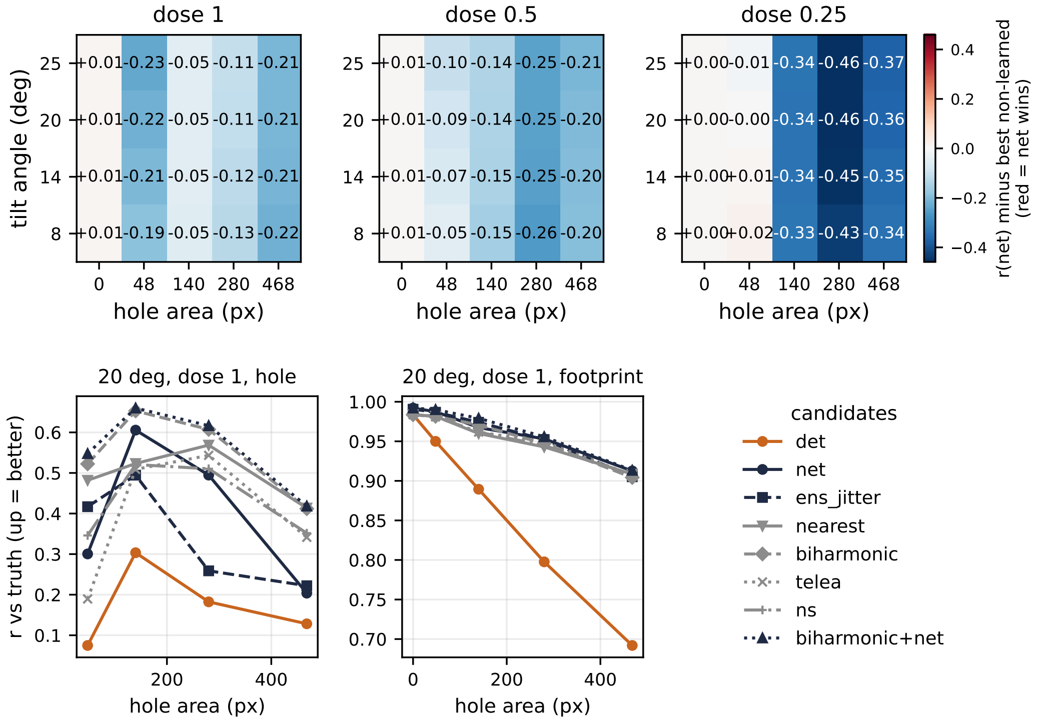

Figure 8: The full regime map: the heat maps of Fig. 4 together with r against hole area at 20°, dose 1, for every candidate, inside the hole and on the footprint (physics orange, learned navy, classical grey, hybrid navy triangles).

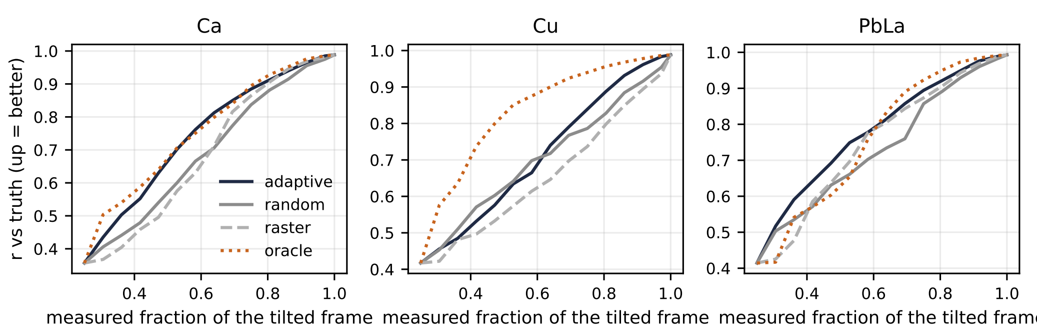

Figure 9: Uncertainty-guided acquisition pilot: r against the measured fraction of the tilted frame when 5×5 tiles are acquired in the order of the jitter-ensemble spread (adaptive), at random, in raster order, or by the true error (oracle, not realisable); 8 tiles per step from a 25 % start, mean of three simulated cases. Adaptive reaches $r \geq 0.90$ (mean over Ca, Cu, Pb Lα) after measuring 79 % of the frame, against 85 % for random or raster order and 72 % for the oracle; for $r \geq 0.95$ the fractions are 88 %, 93 % and 82 %.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The claims of the abstract and introduction (ensemble spread attribution and coverage, defect tolerance, blind identification, the ABC posterior and its ensemble, the regime map) each correspond to a results section (Sections 3 to 6) with the quoted numbers, and the single-instrument, single-painting scope is stated in the abstract and in the Limitations and outlook paragraph that closes Section 6.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: The Limitations and outlook paragraph closing Section 6 states them: one painting and instrument, symmetrised one-sided calibration sigmas, 12-member ensembles, one seed per ladder rung (with a four-seed check on the two borderline rungs), and the resolution limits of rejection ABC; Appendix A adds the seed and error-bar policy.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 2 to 6 define the measured forward model, the knob set with its calibration sigmas, both ensembles, the defect ladder, the diagnostic battery with its null model and pre-registered decision rule, and the ABC protocol; Appendix A gives the data protocol, architecture, optimisation, seed policy and compute.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

   Justification: The code, the calibrated simulator and the per-experiment result tables are prepared for release together with a verification harness, but cannot be linked in a double-blind submission; they will be released with the camera-ready version.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Appendix A specifies the data splits (F1 for training only, F2 and T for evaluation only), the simulation protocol, the architecture, the optimizer, learning rate, batch size, early stopping and the validation set; hyperparameters were fixed once before this study and never tuned on the evaluation data.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

Justification: Every tolerance claim is bounded by the initialisation band of 12 same-seed nominal members (Fig. 2), the two borderline rungs are repeated over four seeds, coverage figures carry standard deviations over lines and cases, and the ABC results are reported at three acceptance fractions.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Appendix A: one 12-core CPU workstation, about 10 minutes per network, 60 trained networks, about 10 CPU-hours in total plus inference-only sweeps; no GPUs were used.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research conforms to the Code of Ethics; it involves no human subjects, no personal data and no high-risk models.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.

- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: The Limitations and outlook paragraph (end of Section 6): the foreseen positive impact is reduced irradiation of heritage objects through uncertainty-guided acquisition; we foresee no direct negative societal impact of restoration methods for elemental maps of paintings.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: No data or models with a high risk of misuse are involved or released.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The classical inpainting baselines are credited through their original publications and the scikit-image paper; the measurement data were acquired by the authors' collaboration and are theirs to release.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The package to be released documents every experiment (per-module contracts and docstrings, a smoke and verification harness, per-experiment summary files); the documentation accompanies the code.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: The paper involves no crowdsourcing and no research with human subjects.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The paper involves no research with human subjects.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [N/A]

    Justification: LLM-based assistants were used for code implementation and text editing under author direction and verification; LLMs are not a component of the methods studied in this paper.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.